NewsBot Intelligence System

INDIVIDUAL REFLECTIVE JOURNAL | IDRIS JUNAID | ITAI 2373

Project overview and individual accountability

The NewsBot Intelligence System integrates the eight NLP areas studied in ITAI 2373 into a practical workflow for classifying and analyzing news articles. I worked with the instructor approved BBC News Classification training dataset, which provides labeled articles in business, entertainment, politics, sport, and technology. This reflection records my verified execution, troubleshooting, evaluation, and documentation activities. It does not claim sole authorship of every component of the group project.

My technical contributions

I obtained the BBC competition training file, resolved an incorrect upload of an unrelated housing dataset, and verified the expected article and category fields. I configured and ran the Colab notebook, examined its preprocessing and linguistic outputs, and checked the integration of TF IDF features, part of speech patterns, dependency relationships, sentiment scoring, multiclass classification, and named entity recognition. I also developed my individual GitHub portfolio documentation and technical report to explain the actual results and system limitations.

Debugging and integration challenges

The most immediate obstacle was environment compatibility. An unrestricted package upgrade introduced conflicting pandas and NumPy versions, followed by a scikit learn import failure. An NLTK VADER resource download was also blocked. I addressed these issues through a fresh Colab runtime, more selective installation, and an alternative VADER package. A later data loading failure exposed the mistaken housing CSV. These experiences reinforced a quality assurance principle: confirm dependency versions, data provenance, schema, and successful execution before interpreting any model output.

Evaluation and observed performance

The completed run retained 1,440 qualifying articles across five categories. A Linear Support Vector Machine was selected after comparing it with Logistic Regression, Multinomial Naive Bayes, and a majority class baseline. Its three fold cross validation macro F1 was 0.9694. On the held out set, it achieved 96.53% accuracy and 0.9645 macro F1. The technology category had the lowest reported recall at about 89%, making category specific error review more informative than accuracy alone. These findings apply to this dataset and split, not automatically to current articles from other publishers.

Reflection, Business Value and Next Steps

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Interpreting outputs beyond classification

The system produced 27,756 named entity mentions and an average of 7.26 extracted syntactic relationships per article, alongside TF IDF, grammatical, sentiment, and entity visualizations. I learned to distinguish an automatically generated signal from a verified fact. Entity cooccurrence does not prove a real world relationship, syntactic patterns do not establish causation, and lexicon based sentiment may miss irony or context. Preserving original capitalization for entities while using normalized text for lexical features illustrated why preprocessing must be matched to the analytical task.

Collaboration and responsibility

The professor framed NewsBot as a collaborative project in which each student maintains an individual portfolio repository. My individual record focuses on the activities I can substantiate from my Colab run, debugging history, results review, and repository preparation. I recognize that integrating a shared project requires crediting the work of teammates and aligning individual documentation with the group deliverable. I have not attributed unverified modules, meetings, or decisions to any teammate. The separate group reflective journal remains a team responsibility under the assignment instructions.

Business value and professional application

NewsBot illustrates how NLP can route articles, surface frequently mentioned organizations or places, and help analysts identify broad coverage patterns. My experience in pharmaceutical quality assurance and biotechnology suggests related exploratory uses in regulatory intelligence, scientific literature monitoring, quality event triage, and risk surveillance. Such systems would require rigorous validation, traceable source data, appropriate access controls, and human review before supporting regulated decisions. The demonstrated model accuracy alone is not evidence of operational benefit or regulatory readiness.

Professional growth and future enhancements

This project strengthened my ability to move from working code to credible analysis. I practiced environment recovery, schema validation, leakage aware model evaluation, clear interpretation of metrics, and reproducible technical communication. A stronger future NewsBot would test newer and out of publisher articles, evaluate per class drift, assess entity and sentiment quality against manually annotated examples, and introduce a review dashboard with correction logs and model version tracking. These steps would make the prototype more accountable and useful in real operational settings.

Conclusion

The central lesson is that an NLP system should be judged by both execution quality and the trustworthiness of its evidence. I successfully executed and reviewed an integrated eight module NewsBot workflow, documented its measured performance, and identified limitations that would matter in real deployment. My individual GitHub portfolio provides the associated notebook, findings, and documentation.

Project repository: https://github.com/IdrisJunaidAI/ITAI2373_NewsBot_Midterm

Source: BBC News Classification, Kaggle (approved course dataset); ITAI 2373 NewsBot midterm instructions; executed NewsBot Colab notebook.